```
1  LOGIC_GND     Logická zem
2  NC   Rezerva
3  NC   Rezerva
4  24V_BATT      Batériová vetva 24 V
5  5V_BATT       Batériová vetva 5 V
6  BATT_GND      Batériová zem
7  5V_LOGIC      LOGIC 5 V
8  LOGIC_GND     Logická zem
9  NC   Rezerva
0  NC   Rezerva
1  I2C_SCL   I²C hodinový signál
2  I2C_SDA   I²C dátový signál
3  NC   Rezerva pre budúce IRQ
4  NC   Rezerva
5  SPI_SCK   SPI clock
6  LOGIC_GND     Referenčná zem SPI
7  SPI_MOSI      SPI controller → periféria
8  SPI_MISO      SPI periféria → controller
9  SPI_CS0   Výber SPI zariadenia 0
0  SPI_CS1   Výber SPI zariadenia 1
1  SPI_DC    Display Data/Command
2  LOGIC_GND     Referenčná zem SPI
3  NC   Rezerva
4  RS485_GND     Referenčná zem RS-485
5  RS485_A   Diferenciálna linka A
6  RS485_B   Diferenciálna linka B
7  NC   Rezerva
8  NC   Rezerva
```

BACKBONE V2.1
Medulla Oblongata

H101
M20-6102045

24V, GND, 5V
LOGIC_GND
24V_BATT, BATT_GND, 5V_LOGIC
I2C_SCL, I2C_SDA
SPI_SCK, SPI_MOSI, SPI_SC1, SPI_DC, UART_TX, RS485_A, UART_RX
5V_BATT, BATT_GND, LOGIC_GND
SPI_MISO, SPI_CS2, LOGIC_GND, RS485_GND, RS485_B, LOGIC_GND

R120
MBB02070Z0000ZC100
24V_LOGIC

MANUS

IN LINE PULL UP ULNA

R102 ERJPA3F4701V — LOGIC_3V3 — POLLEX
R103 ERJ-3EKF1001V
C102 C0603C103K5RACTU — LOGIC_GND

R104 ERJPA3F4701V — LOGIC_3V3 — INDEX
R105 ERJ-3EKF1001V
C103 C0603C103K5RACTU — LOGIC_GND

R106 ERJPA3F4701V — LOGIC_3V3 — DIGITUS_MEDIUS
R107 ERJ-3EKF1001V
C104 C0603C103K5RACTU — LOGIC_GND

R108 ERJPA3F4701V — LOGIC_3V3 — DIGITUS_ANULARIS
R109 ERJ-3EKF1001V
C105 C0603C103K5RACTU — LOGIC_GND

CN101 1990067
LOGIC_GND ×8

DIGITUS MINIMUS

0-30V OUT LINE RADIUS

Pull-down na GPIO
R112 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R110 ERJ3EKF3300V
OS_Trapenzium
U101 AQY211EH
LOGIC_GND
D101 SMBJ26CA_C83332

Pull-down na GPIO
R111 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R113 ERJ3EKF3300V
OS_Trapezoideum
U102 AQY211EH
LOGIC_GND
D102 SMBJ26CA_C83332

Pull-down na GPIO
R114 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R115 ERJ3EKF3300V
OS_Capitatum
U104 AQY211EH
LOGIC_GND
D103 SMBJ26CA_C83332

Pull-down na GPIO
R116 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R117 ERJ3EKF3300V
OS_Hamatum
U105 AQY211EH
LOGIC_GND
D104 SMBJ26CA_C83332

CN102 1990067
LOGIC_GND ×8

ESP32-S3-WROOM-1U-N16R8

ADC1, ADC2
DI1_LOGIC POLLEX
DI2_LOGIC INDEX
DI3_LOGIC DIGITUS_MEDIUS
DI4_LOGIC DIGITUS_ANULARIS
DI_AUX1
DO1_CMD OS_Trapenzium
DO2_CMD OS_Trapezoideum
DO3_CMD OS_Capitatum
DO4_CMD OS_Hamatum
DO_FAULT_INT
ESP_BOOT
RS485_A, RS485_B
LOGIC_GND, LOGIC_3V3
I2C_SCL, I2C_SDA
JTAG TDI, JTAG TMS
RGB_DATA
SPI_CS1, SPI_CS2, SPI_MISO, SPI_MOSI, SPI_SCK, SPI_DC
UART0_RX, UART0_TX
UART_RX, UART_TX
CHASIS

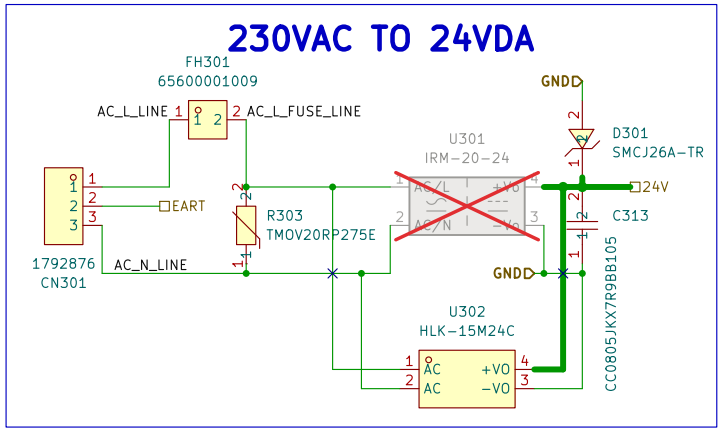
R121
MBB02070Z0000ZC100
LOGIC_3V3

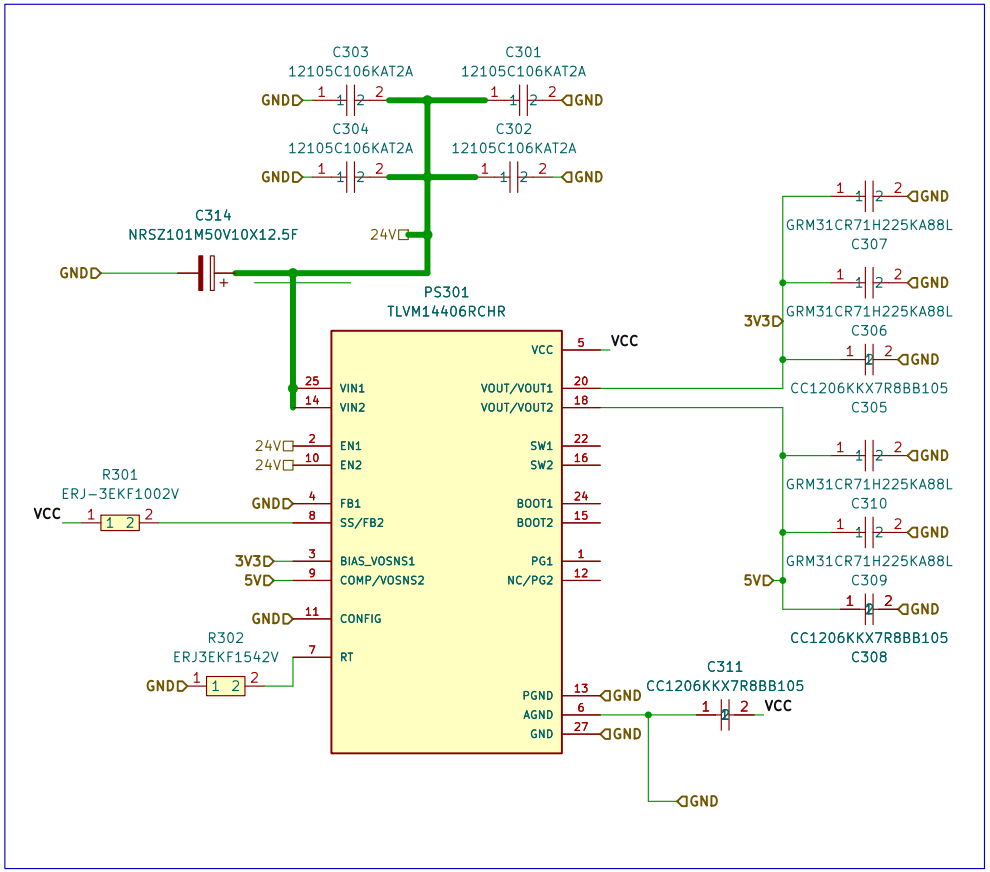
36V_TO_5V+3V3
LOGIC_3V3 → 3V3
R119
5V_LOGIC → 5V
MBB02070Z0000ZC100
24V_LOGIC → 24V
CHASSIS_GND → EART
LOGIC_GND → GND

RGB

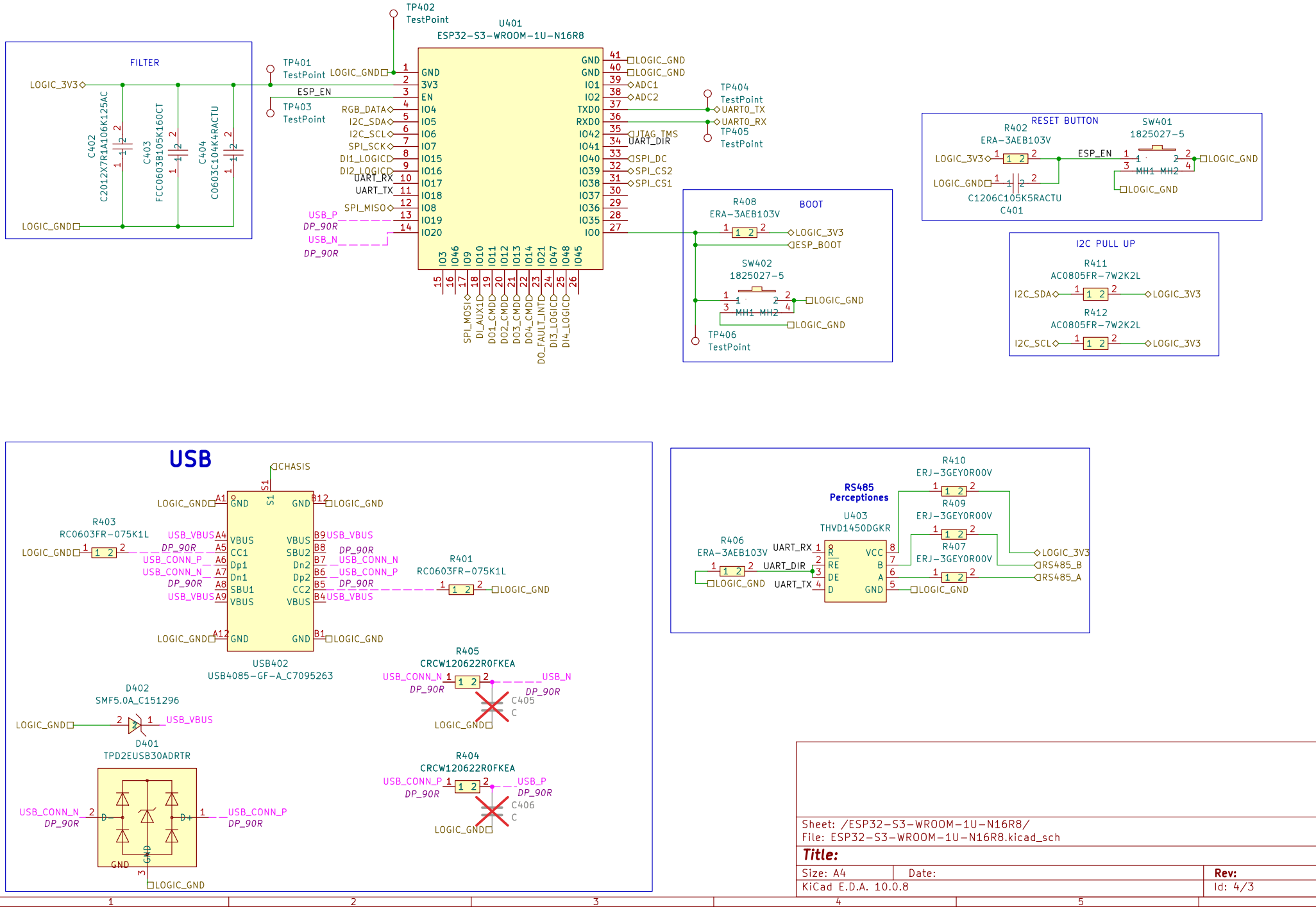
U103
SN74AHCT1G125DBVR
OE, A, GND, VCC, Y
LOGIC_GND
C101 C0805C104K5RACTU_C3018562
RGB_DATA
R118 ERA-3AEB103V — LOGIC_GND
R101 ERJ3EKF3300V
5V_LOGIC
CN104 1989751

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Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 1/3





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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/3


FILTER
LOGIC_3V3
LOGIC_GND
C402
C2012X7R1A106K125AC
C403
FCC0603B105K160CT
C404
C0603C104K4RACTU
TP402
TestPoint
U401
ESP32-S3-WROOM-1U-N16R8
TP401
TestPoint
TP403
TestPoint
ESP_EN
RGB_DATA
I2C_SDA
I2C_SCL
SPI_SCK
DI1_LOGICD
DI2_LOGICD
UART_RX
UART_TX
SPI_MISO
USB_P
DP_90R
USB_N
DP_90R
GND
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19
IO20
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
SPI_MOSI
DI_AUX1D
DO1_CMDD
DO2_CMDD
DO3_CMDD
DO4_CMDD
DO_FAULT_INTD
DI3_LOGICD
DI4_LOGICD
IO1
IO2
TXD0
RXD0
IO42
IO41
IO40
IO39
IO38
IO37
IO36
IO35
IO0
ADC1
ADC2
UART_DIR
SPI_DC
SPI_CS2
SPI_CS1
JTAG_TMS
TP404
TestPoint
UART0_TX
UART0_RX
TP405
TestPoint
R408
ERA-3AEB103V
BOOT
LOGIC_3V3
ESP_BOOT
SW402
1825027-5
MH1
MH2
TP406
TestPoint
RESET BUTTON
R402
ERA-3AEB103V
SW401
1825027-5
C1206C105K5RACTU
C401
I2C PULL UP
R411
AC0805FR-7W2K2L
R412
AC0805FR-7W2K2L
USB
CHASIS
S1
USB402
USB4085-GF-A_C7095263
R403
RC0603FR-075K1L
R401
RC0603FR-075K1L
D402
SMF5.0A_C151296
D401
TPD2EUSB30ADRTR
R405
CRCW120622R0FKEA
C405
R404
CRCW120622R0FKEA
C406
RS485
Perceptiones
U403
THVD1450DGKR
R406
ERA-3AEB103V
R410
ERJ-3GEY0R00V
R409
ERJ-3GEY0R00V
R407
ERJ-3GEY0R00V
RS485_B
RS485_A
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 4/3